MUHAMAD ILHAM FADILLAH SUHERLAN

SOFTWARE ENGINEERING FRESH GRADUATE | SYSTEM ANALYST / BACKEND DEVELOPER / MOBILE DEVELOPER
Cimahi, Indonesia | +6281214145706 | muh.ilham.fs@gmail.com
http://www.linkedin.com/in/muhamadilhamfs | https://github.com/ilhamfs7 | https://muhilhamfs.vercel.app/

SUMMARY

Software Engineering fresh graduate with hands-on internship and academic project experience across the software development lifecycle, strongest in requirements analysis, system and database modeling, and technical documentation. Experienced in translating stakeholder needs into functional and non-functional requirements and carrying them through design, implementation, testing, and AWS deployment, with practical experience in backend development (PHP/Laravel, PostgreSQL, REST API) and mobile development (Flutter, Kotlin).

EDUCATION

Diploma (D3) in Software Engineering | Politeknik Negeri Bandung, Indonesia | GPA 3.58/4.00 Aug 2023 – Aug 2026

TECHNICAL SKILLS

- **Requirements & System Analysis:** Requirements Gathering & Elicitation, Stakeholder Interview, Existing-System Analysis, Business Rules, Functional & Non-Functional Requirements, Use Case Analysis, Requirements Validation
- **System Modeling & Documentation:** UML (Use Case, Sequence, Class, Package, Deployment), Domain Model, DFD, ERD, Conceptual/Logical/Physical Data Model, SRS, System Design Documentation, User Guide
- **Backend & Database:** PHP, Laravel, Java, Spring Boot, REST API, SQL, PostgreSQL, Database Design
- **Mobile Development:** Flutter, Dart, Kotlin, Jetpack Compose, MVVM, API Integration
- **Testing:** Test Case Development, UAT, Integration Testing, Black-box Testing, Validation Testing
- **Cloud & Tools:** AWS (EC2, S3, RDS, SQS), Git, GitHub, Postman, Figma, Draw.io, PlantUML, PowerDesigner
- **Programming Languages:** Python, C, C++

WORK EXPERIENCE

Balai Bahasa Provinsi Jawa Barat | Bandung, Jawa Barat Jul 2025 – Oct 2025
Intern – System Analyst & Mobile Developer | Flutter, Dart, MVVM

- Conducted stakeholder interviews and existing-system analysis; translated 2 use cases and 3 sub-use cases into 9 functional and 4 non-functional requirements.
- Designed the ERD and UI, and wrote the user guide and manual for Kado Ceria, a bilingual (Sundanese–Indonesian) interactive audio storybook app for early childhood, BIPA learners, and persons with disabilities.
- Implemented the main flipbook feature using Flutter/Dart (MVVM), and demonstrated the app through UAT and a handover presentation at the institution.

PROJECT EXPERIENCE

CONTENT MANAGEMENT SYSTEM – KADO CERIA | github.com/DAliRIJTK/cms-kadoceria Jan 2026 – Aug 2026
System Analyst & Technical Writer | Project Lead | Backend Developer | QA | Laravel, PHP, PostgreSQL, REST API, AWS

- Analyzed the existing Kado Ceria mobile app, where every flipbook content update required code changes and redeployment, and interviewed Balai Bahasa Provinsi Jawa Barat to define content management requirements.
- Primary author of the SRS: elicited 10 user requirements and 14 business rules, elaborated into 34 functional and 10 non-functional requirements across 2 use cases.
- Documented system design (architecture, package diagram, 12 sequence diagrams, class diagram, deployment diagram) and conceptual/logical/physical data models for the PostgreSQL database.
- Prepared 44 UAT test cases and validated requirements through integration (15) and black-box/validation (43) testing, each with a 100% pass rate.
- Fixed bugs affecting audio features, book conversion, and book CRUD in the existing Laravel/PHP codebase, and refactored background audio logic to reduce redundant code.
- Deployed the completed system to AWS using EC2, S3, RDS, and SQS.
- Led the project team, including task allocation and scheduling.

MOBILE APPLICATION – GROW | github.com/grow-app-4/grow-app-mobile Feb 2025 – Jun 2025
System Analyst | Mobile Developer | Kotlin, Jetpack Compose, MVVM, PostgreSQL

- Identified stakeholders, designed the domain model, and translated 5 use cases and 1 sub-use case into 9 functional and 7 non-functional requirements.
- Designed the ERD and PostgreSQL database; implemented the daily intake monitoring feature (mother's intake and child's breast milk intake) with local storage and API integration; developed 10 test cases (100% pass).

ADDITIONAL TECHNICAL PROJECTS

- **Suku Face Recognition (Python)** – refactored preprocessing using MTCNN Feb 2025 – Jun 2025
- **Gamification - Semaphore (Python)** – modified detection logic and input vs target comparison Feb 2025 – Jun 2025
- **Hidingdata (C)** – implemented LSB encryption/decryption with a doubly circular linked list Feb 2024 – Jun 2024
- **Search Application Using Trie Data Structure (C++)** – implemented Trie-based search Feb 2024 – Jun 2024

CERTIFICATION & TRAINING

- **Arutala Junior Backend Developer using Spring Boot – Distinguished** | Arutalalab July 2026
- **Test of English for International Communication (TOEIC)** | Score: 765 May 2026
- **Basic Management Skills Training (LKMM-TD)** | Politeknik Negeri Bandung Nov 2023